

Using a Virtual Machine to Run Linux on Windows

If you would like to run Linux on Windows and have never used a virtual machine, this tutorial explains how you can do so in a few simple steps.



Linux and Windows are two of the most popular operating systems in the market. One can say that these two OSs have become essentials because of their continuous development and feature enhancements. Many people prefer one over the other. Yet, people who use Linux sometimes need the Windows operating system, and vice versa. For faster development and more security, people prefer to use Linux over Windows. So these people need Linux alongside Windows.

A common solution for this problem is to use a dual boot, which basically installs the entire operating system. However, due to boot loader changes (like UEFI mode), configuring dual boot becomes difficult.

This article is about running Linux on Windows, without using the dual boot technique. There are different ways to do so, one of the most popular being using a virtual machine.

Virtual machines

Virtualisation is fairly old technology now. It was started around 1980 by IBM. So, currently, this technology is pretty stable. We can create Linux virtual machines in Windows using software like Virtual Box, VMware Player, and VMware Workstation.

A virtual machine provides the same functionality as a physical computer, but it is an emulation of a physical computer. The virtual machine can be created or destroyed at any moment and it will not impact the actual operating system, which is the best part about using it.

To install Linux on Windows, we just require two things — the software to create a virtual machine which is Virtual Box (free software) and an ISO file of the operating system.

The installation process is pretty simple, like installing any software in Windows. The steps for installing Ubuntu (a favourite Linux flavour) inside Windows are given below.

The first step is to enable virtualisation technology in the BIOS. If you have already enabled it, you can skip this step.

You need to power on the system and go to the BIOS setup by pressing F2, Del or the Enter key, depending upon your computer model.

After that, under *CPU Configuration* -> *System Configuration* -> *Advanced* or *Security Tab*, we need to find *Virtualization Technology* or *Intel Virtualization Technology*. If the option is disabled, then it must be enabled. If it's already enabled, skip this step.

If you get any of the following error messages at any point